

grandMA2 for Eos Programmers

and vice versa

A guide for whether you're in a pinch or just want to learn...

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Introduction

This guide started as a small note on my phone several years ago, when I decided to start learning the **grandMA** on my own. I had become proficient in using **Eos** consoles, but felt like I was completely starting over when it came to learning **grandMA**.

I began scouring the online manual, watching tutorials, and just playing around with **onPC**. I slowly started to realize that the two consoles weren't really all that different at their core. Many of the concepts and actions are extremely similar, just using different words and syntax. I started to keep notes on common **Eos** commands and their **grandMA2** equivalent, and suddenly that big 'ole **grandMA2** didn't look nearly as intimidating as it did before. I discovered that nearly all of the functions in **Eos** can be accomplished on the **grandMA2** and vice versa (as well as a majority of other popular consoles). It's just a matter of knowing the words and commands to use and in which order.

Because this guide was written from such a perspective, the majority of the comparisons and descriptions are more geared towards those going from **Eos** to **grandMA2**. However, this is not to say that the guide couldn't be used in the opposite manner as well, though some of the explanations and examples may be less thorough.

It is assumed that the reader/programmer has a general knowledge and understanding of the basic philosophies of lighting control, including:

- Tracking vs Cue Only
- LTP (Latest Takes Precedence) vs HTP (Highest Takes Precedence)
- Priorities and Background states
- Referenced Data vs. Absolute Data
- Basic functions of **ETC Eos** consoles (or **grandMA2** consoles).

While the guide is loosely structured in a way that can be read start to finish as a full tutorial, it is probably more useful as a reference.

As such, you are strongly encouraged to use your device's find/search function to search for whatever terms or commands you are looking for, either **Eos or **MA2**.**

Tips for reading the guide

Orange = gMA2 command, key, or term (i.e. "Please")

Blue = Eos command, key, or term (i.e. "Enter")

{key combo or softkey} = some commands are accessed with a combination of key presses.

|| = Used to separate multiple commands in sequence.

i.e. "Blind || Preset 1 thru 12 || Chan 4 At 30"

i.e. "Fixture 34 At Preset 0.13 || Park Fixture 34"

A few core concepts about MA2:

SYNTAX ORDER

In general (though not necessarily true for all commands) MA2 operates with the syntax of [Action] [Object/Target] At [Object/Target]

Whereas Eos generally operates (though also not true for all commands) with the syntax [Object/Target] [Action] [Object/Target]

For Example:

"Move Preset 4.2 At 4.3" = "Color Palette 2 Move To Color Palette 3"

"Label Cue 2" = "Cue 2 Label"

[Action] [Target] = [Target] [Action]

Don't worry if this doesn't make sense immediately, and don't get caught up on it as some commands don't always follow this structure. Just remember that if a command you're trying on MA2 isn't working like you'd expect on Eos, try reversing the syntax order.

CHANNEL SELECTION

On MA2, channel selection is independent of the command line.

Meaning once you've selected channels, the selection will remain regardless of what is typed into the command line until you make a new selection or clear the selection (Clear key one tap).

That is not to say, however, that channel/fixture selections cannot be apart of SOME commands.

i.e. "Fixture 34 At Preset 4.34" is the equivalent of making a new selection AND specifying a new value.

In general, channel selections are NOT required for most commands because again channel selection is independent of command line and is assumed as the target of most commands until otherwise cleared or specified.

On **Eos** every channel-related command requires a channel selection to accompany it. Depending on the order of operations, the software will automatically input the previous selection to the command line (i.e pressing **At** after terminated command re-types previous selection), making it fairly easy and quick to type new commands. And thus the extremely important and useful command of **Select Last**.

- Selection Syntax: "**Fixture 304 Please**" = "**Chan 304 Enter**" = Channels/Fixtures Selected
 - Then "**At 40 Please**" = "**Select Last At 40 Enter**"
-

THE "PROGRAMMER"

The "**Programmer**" is the term used to describe the source of all **Manual** data (anything in red). If a fixture or attribute is in the "**programmer**" it means it is being manually controlled by commands (not stored in any cue/playback/sequence/executor)

There are two "states" of manual programmer values: **Active** and **DeActive**.

Active means "**Manual**" **and** available for record/store
(In **Fixture Sheet**: red background and red text)

DeActive also means "**Manual**" but **not** available for record/store (like "**make null**" in **Live**)
(In **Fixture Sheet**: only red text, no red background)

So if all of your current manual values in the programmer are **DeActive**, and you try to **Store** a cue or preset, nothing would actually be stored because the values are not **Active**.

Activate {**Please Please** or **double tap group** or **Preset Type**} is essentially "**Make Manual**" + making available for store actions
{**Please Please** or **double tap group** or **preset type**} a 2nd time will **DeActivate**, but will remain in the programmer.

Both active and inactive channels/fixtures come from the "**Programmer**"

By default, any **Active** data will become **DeActive** after a store command. Meaning that if you wish to store the data again, you must re-**Activate** it. (These are all settings that can be changed in **Store Options**)

FIXTURE # VS CHANNEL

gMA2 allows for both a Fixture # and Channel # simultaneously for every patch object. This does **NOT** delineate what type of object can be patched to either number. Meaning any object can contain one or the other or both.

i.e. A moving light could be given a Channel #, and a simple dimmer could be given a Fixture #, or vice versa.

The only delineations these two numbers make is whether the object shows up in the **Channel Sheet**, the **Fixture Sheet**, or both, and also whether the object is accessible via the {**Fixture**} key, the {**Channel**} key, or both.

i.e. Fixture 20 and Channel 20 **could** mean two different separate objects.

For the most part, however, many programmers keep these numbers the same for all patch objects, or use just one or the other, to avoid confusion. So for example, your 20 Mac Vipers could have Fixture #'s of 1-20, as well as Channel #'s of 1-20. Or just Fixture #'s only and no Channel #'s.

But, this can also open up a lot of customization as well. For example, to delineate between simple dimmers (Channels 1-24) vs moving lights (Fixtures 1-10). Or separate channeling schemes for designer and programmer to reference. Or to easily delineate what objects show up on the fixture sheet vs the channel sheet.

ORGANIZATION OF ATTRIBUTES

There are 3 levels of organization for attributes of a Fixture/Channel

PresetType -> Feature -> Attribute

PresetType $\cong$ Encoder Group

Feature $\cong$ Encoder Group Page

Attribute = Attribute/Parameter

These are all keywords which can be used as part of a command.

i.e. "Off PresetType 2"

i.e. "On Attribute "Dim""

i.e. "Feature 1.2 At 50"

All PresetTypes, Features, and Attributes can be identified either with numbers (PresetType 2 = PresetType "Position") or with words in quotes (Attribute "Dim" specifies the Dimmer Attribute)

PresetType {MA + Preset}

Feature (not accessible via hard keys)

Attribute {Preset Preset}

You can also tap any of these objects on the Encoder Bar in place of typing it out specifically in a command line.

i.e. Tap on the "Position" Preset Button in encoder bar = "If PresetType "Position""

http://help2.malighting.com/Page/grandMA2/keyword_presettype/en/3.3/presettype

http://help2.malighting.com/Page/grandMA2/ws_preset_control_bar/en/3.3

A NOTE ON WORKFLOW AND COMMANDS

It can be useful to understand from the beginning that MA2 is generally designed to be very touch-screen oriented. So while the MA2 is still a command-line based console like the Eos, the workflow on the MA2 tends to spend much less time on the keys, and more time on the touchscreens.

i.e. Moving Presets or other pool objects is usually accomplished by pressing the Move Key, tapping the object you want to move, and then tapping where you want to move it to.

******For the remainder of this guide, anytime any Object/Target is specified within a command , it can generally be assumed that you can also tap that specific pool object in place of typing out the Object/Target with the keys.

Conversely, all commands in MA can also be executed *without* any hard key console presses, by simply typing out the command structure via the standard keyboard on your computer or under the armrest of the console. This is **NOT** possible on the Eos console.

In fact, there are many keywords/commands within MA that are actually not even accessible via HardKeys, and must be typed out on the keyboard.

GENERAL COMMANDS AND TERMS:

Please = Enter

A Please or Enter is required after every command in order to execute it
(From here on out unless otherwise noted, all example commands assume that there is a Please or Enter at the end)

(|| will also indicate a Please or Enter)

Oops is Undo

Clear does **NOT** behave the same way as Clear in Eos

{Clear} = ClearSelection - Clears out the current channel/fixture selection

{Clear Clear} = ClearActive - DeActivates all fixture/channel data currently in Programmer

{Clear Clear Clear} = ClearAll = Sneak Enter - Removes all manual data from Programmer

Escape is like Clear in regards to command lines

Group = Group

Store is essentially Record...

However grandMA is, by default, always in a "Record Only" state. It will only store data that you've manipulated or Activated in the Programmer, meaning it won't store values that are already coming from a cue or executor (unless you specifically say otherwise in the "store options menu". i.e. There is an option to store all output regardless of whether it is manual/ from the programmer)

Press and hold Store to see Store Options pop up

So, by default, Store is actually more like Record Only

At = At for setting intensity or other preset data

"Fixture 205 At 60" = "Chan 205 At 60" = Sets intensity to 60

"Fixture 205 At Preset 0.65" = "Chan 205 At Preset 65"

At also = Recall From.

"Fixture 201 At Fixture 202" = "Chan 201 Recall From Chan 202"

"Fixture 201 At Cue 44" = "Chan 201 Recall From Cue 44"

******See section on Status to include tracked values in your command, as MA2 by default will **NOT** include tracked values when pulling data from other Cues.

{At At} is like "At Level", the level of which can be specified in Setup. Default is Full.
So {At At} brings current selection to Full

".." or "{dot}{dot}" is like "Out" – So ".." brings current selection to 0

DMX {Channel Channel} is equivalent of Address
"DMX 4.50 At 75" = "Address 4/50 At 75"

Full = Full
(wouldn't it be funny if it wasn't?)

DISPLAYS & VIEWS

{Tap on empty screen space} = {New Tab}

Pools are essentially the equivalent of Direct Selects

(Though, Pools can also be set to display in a list view, similar to Eos Target lists (i.e. "{Preset Preset}" brings up list view of Presets)

Click/Tap on the yellow ball in upper left corner of a Pool to see all Pool options (and to set the pool to display in list format)

Command Line = Command History

MA2 Command Line also shows errors and whether a command was successful or not. Very useful.

"Fixture Sheet" is essentially "Live Table" Tab (Will only show objects with Fixture # in patch)

"Channel Sheet" is sort of like "Live Tombstone" Tab (Will only show objects with Channel # in patch)

"Sequence Executor Sheet" is essentially "Playback Status Display" Tab

"Sequence Tracking Sheet" is similar to a "Blind Spreadsheet" tab in Eos

Views are Snapshots (but only in the sense of displays, nothing to do with filters or Fader status)

{Setup} is combination of {Displays Setup} and the Eos Settings Shell (for networking, system time, ...). Almost all settings and setup can be found in {Setup} screen

- In Eos, all networking setup occurs outside of the main programming environment.

{Displays} Scroll down to Exit. Then click on Settings

{Backup} is for all show file Save and Open functions

{Backup}{Backup} = {Shift Update} - Quick Save

PATCH

Patch is found in the {Setup} menu.

Be aware of both Fixture # and Channel #, though in many instances they are kept identical, or only one or the other is used. (See section on "Fixture # vs Channel #" under Core Concepts)

MultiPatch = approximately **Channel Parts** = Patching multiple addresses to one Channel/Fixture

(i.e. To patch addresses 1 thru 10 to channel 1, create channel 1, then press the softkey **Create MultiPatch**)

***MultiPatch** can only be created for the same Fixture Type i.e. a dimmer and an iris can **not** be in the same MultiPatch.

*An example of this would be an I-cue or Auto-Yoke or VL-1000 with external dimmer.

In **Eos** these fixtures can be patched simply by breaking down the separate addresses and types into **Channel Parts** and then patching the starting **addresses**.

In **MA2**, you must create or use a specific **Fixture Type** that specifies all of those individual parts. In the **Fixture Type** editor, the **Break** column, defines the separation between chunks of DMX addresses.

So for a typical I-Cue fixture setup, the Dimmer would be within **Break 1**, the I-Cue would be within **Break 2**, and the Iris would be within **Break 3**. Then when patching, you are given the option to assign a separate starting address for all 3 of these "Breaks"

Patch syntax via command line (Fixture Objects must already exist in the Patch Schedule):

"Assign Fixture 201 At DMX {Channel Channel} 4.101" = "Patch || Chan 201 At 4/101"

Overwrites previous address assigned to fixture.

"Assign DMX {Channel Channel} 4.101 At Fixture 201" = "Patch || Chan 201 Part # At 4/101"

ADDS address to channel as **MultiPatch**.

MORE COMMANDS

Next & **Last** partially behave the same way as **Next** & **Last** in Eos, but **only** in terms of channel selection.

i.e. "Group 18 Please" {Next}{Next}{Next} = "Group 18 Enter" {Next}{Next}{Next}

Next & **Last** are used in Channel Selection, **MAtricks**, **Highlight**, **Solo**

****MA2 Next** & **Last** will **NOT** function with any other type of object or target, other than channel selection.

Whereas Eos **Next** & **Last** can pertain to whatever object or target is currently in the command line.

A **Sequence** is a **Cue List**

Executor is **Fader** on Eos (as of Eos v2.4).

An **Executor** contains a **Sequence**. A **Sequence** contains **Cues**.

Select acts sort of like the **Master Load** button

i.e. "Select Executor 1.14" = "Cue 14/ [Load]"

(Select also can be used with other targets such as **Filters**)

Break Cue is a **Block Cue**. (This is found in the **Sequence Executor Sheet** under **Cue Type**)

Block and **Unblock** are also keywords that can be used for **forcing tracked values to be hard values**, either per cue or per fixture.

"Block Cue 4" or "Block Cue 4 If Fixture 34"

But these commands only apply to existing values or tracked values in a cue, and act only as a one-time action. If a new fixture is added to the sequence later on, only a **Break Cue** will block the newly tracked values.

MIB (Move in Black) is like "**Marking**". Though it is a little more like **Eos AutoMark**, in that in **MA2** you cannot specify specific Channels for Marking/MIB. You can only apply MIB options on a per Cue basis.

* in the **MIB** column of the **Sequence Executor Sheet** is essentially the same as a **Live Move Flag** in the **Playback Status Display**.

In general, it is very time consuming to manually **MIB** cues, so it's usually easiest to just leave it on Auto in the **Executor Options**.

Masks are like **Flexi** (but much more customizable and complex).

Masks are used in the **Fixture Sheet** and **Channel Sheet** to “mask” away different fixtures/channels.

Filters {**Group Group Group**} are like **record filters/snapshots** or **selective recording**.

Press and hold **At** to open the temporary command filter. This means that the next command you type/execute will only apply to the attributes that you select in this command filter screen). You can then store your current attribute selection as a **Filter** object for later use if you wish.

Press “**Edit Filter #**” to create a new **Filter** in the Filter pool.

“**Filter 3**” or simply tapping a **Filter** Pool tile will only **temporarily** activate the filter for the **next command only**.

“**Select Filter 3**” will permanently change the Filter for all future commands until a new Filter is selected.

PRESETS

Preset is like a combination of **Palettes** and **Presets**.

i.e. **Color Preset, Position Preset, Gobo Preset** = **Color Palette, Focus Palette, Beam Palette**. (Only that category of attribute can be stored to that preset)

An "**All-Preset**" is the equivalent of "**Preset**" on Eos, in that all attributes can be stored to this type of preset.

****Technically, any preset in MA2 can store any attribute info if the "Preset Filter" in Store Options is turned off. But by default, the Preset Filter is on and presets will only store the attributes of their PresetType.**

****There is a setting called "Allow Embedded" in Store Options, which is essentially the equivalent of being able to store Palettes within Presets. It will allow Presets to be stored within other Presets.**

PresetType is the category of gMA2 Preset (Color, Position, etc.)

(a term used often in MA language, which is why I mention it)

Each **PresetType** has an equivalent number (Dimmer=1, Position=2, Color=4, All=0, etc.) to specify PresetType objects within the command line.

So i.e. "**Preset 4.34**" refers to Color Preset 34, "**Preset 2.7**" refers to Position Preset 7

i.e. "**At Preset 4.34**" will place the currently selected fixtures into Color Preset 34.

The encoder grouping touchscreen buttons above the encoders are called the **PresetType** buttons. These can be used in combination with many other commands such as **Off, On, Extract, Default, Release, Remove, If**, etc.

i.e. "**Off (Touch "Position" PresetType Button)**"

Preset Store/Scope options:

Selective is like **discrete palette** data

Global is **By-Type**

Universal is a super special **By-Type** that will apply to any and all fixtures with those parameters. Only works for intensity, position, and mix color.

Found in **Store Options** or **Preset Pool** window options.

IF

IfOutput {If} is **select active**

IfActive {If If} is **select manual**, but doesn't include deactivated (red text only) data

IfProg {If If If} is **select manual**, both active and deactive (red text and red background)

IfOutput, IfActive, and IfProg also work very similarly to **Query** when used in combination with **At** and with other targets/objects.

"IfOutput At Preset 4.34" = "Query Color Palette 34"

"IfOutput At 0" = "Query At 0"

If {MA + If} or {If If If If} is primarily used as a helping keyword with many other commands to limit the scope of that command.

For example **Selective Recording**:

"Store Cue 34 If Fixture 9" = "Chan 9 Record Cue 34"

"Store Cue 34 If PresetType 4" = "Color Record Cue 34"

Or recalling data from other objects or sources:

"Fixture 202 At Fixture 203 If PresetType "Color"" = "Chan 202 Color Recall From Chan 203"

"Fixture 41 At Cue 23 If Attribute "Pan"" = "Chan 41 Pan Recall From Cue 23"

"Block Cue 34 If Fixture 9" = "Blind || Cue 34 || Chan 9 Block"

OTHER COMMANDS

Copy is essentially the same as **Copy To**, but requires different syntax.

"Copy Preset 4.2 At 4.3" = "Color Palette 2 Copy To Color Palette 3"

"Copy Cue 4 At Cue 22" = "Cue 4 Copy To Cue 22"

See section on **Status to include tracked values in your Copy command

Move is essentially the same as **Move To**, but requires different syntax.

"Move Preset 4.2 At 4.3" = "Color Palette 2 Move To Color Palette 3"

"Move Cue 4 At Cue 5" = "Cue 4 Move To Cue 5"

MAtricks is **Offset**

Wings is like **Pan Fan Mirror** (Think of symmetry of "wings" on a bird or plane)

Single X is simply the individual selection within a group that is controlled by {**Next**} & {**Last**}

Blocks are like {**Chan Per Group**}

Interleave is like {**Chan Per Group**} {**Interleave**}

<http://help2.malighting.com/Page/grandMA2/matricks/en/3.3>

Align is **Fan**

You can also use the syntax "At 10 thru 100", much the same as you would in **Eos** to **Fan** values.

(Note: This can **NOT** be done with Presets or other referenced data on **gMA2**, as you can in **Eos**. Only possible with hard values on **gMA2**.

i.e. "At Preset 4.1 thru 4.5" does **NOT** currently work)

{**Assign Assign**} is **Label** - i.e. "Assign Assign Cue 2 Please" opens dialog to label Cue 2 (Eos syntax: "Cue 2 Label")

Worlds {**Group Group**} are like **Partitions**, but way more flexible, in that you can also exclude/include specific attributes.

Worlds are like a combination of **Partitions** and **Record Filter Snapshots**.

Stomp {**MA+At**} is like "Stop Effect" command, but if applied to an attribute or Fixture acts more like a value than a command (see section on **Effects**) and behaves as such in regards to LTP, higher/lower priorities, and background values.

i.e. "Stomp Selection" or "Stomp PresetType "Dimmer"" will stop any running effects respectively in the **Programmer**

****Not to be confused with Eos Stomp (v2.4)** Eos Stomp has to do with a Fader (cue list, submaster, etc) being overwritten by another Fader (cue list, submaster, etc). The Eos Fader options for Stomp dictate what the Fader will do when it no longer "owns" the fixtures or attributes, when another Fader has taken over control of those fixtures. See *Executor Options* for gMA2 equivalent.

Layout is like a Magic Sheet

To put Fixtures/Channels in a Layout, make a selection then "Store Layout #". Those fixture objects will then appear in the Layout.

To place any other object (Presets, Groups, Macros): "Assign Preset # At Layout #". Those objects will now appear in the Layout.

Press Setup at the top of the Layout View to arrange your Layout objects.

Freeze is capture

Freeze on MA is just an on/off state of the programmer. It cannot be applied to individual fixtures/channels.

Call {On On} is the equivalent of "Recall From [Record Target]", meaning it will put all stored data of that object into the Programmer.

i.e. "Call Preset 4.2" = "Recall From Color Palette 2"

OFF & ON

On is like Make Manual (and also Activates), in regard to Fixtures or Attributes or PresetTypes

i.e. "On Fixture 201" = "Chan 201 Make Manual"

i.e. "On Selection {Fixture Fixture}" = "Select Last Make Manual"

i.e. "On {Touch Color PresetType encoder group}" = "Select Last Color Make Manual"

Off is like Sneak, in regards to channels or attributes

i.e. "Off Fixture 201" = "Chan 201 Sneak"

i.e. "Off Selection {Fixture Fixture}" = "Select Last Sneak"

i.e. "Off {Touch Color PresetType encoder group}" = "Select Last Color Sneak"

Both of these also have functions relating to executors and playback that is more like releasing content and "Go To Cue 0" and such

i.e. "Off Sequence 12" is somewhat equivalent to "Go to Cue 12/0"

(gMA2 "Off" is also essentially equivalent to Eos "Off", in that gMA2 will release control of that Sequences/Executors fixtures to the background values, just like in Eos)

"Off Page Thru" = "Go To Cue Out", in that it is turning off all executors on all pages.

"Off Please" = "Go To Cue 0" will turn off the currently selected executor

"Off Executor 3" = "Go To Cue 3/0"

Off Off opens up the Off Window, which shows you all running executors, effects, macros, etc.

Off and On are very useful and powerful commands

"Selection" {Fixture Fixture} - makes the current channel selection usable in a command.

i.e. "Off Selection" = "Select Last Sneak" - will knock out the currently selected channels from the programmer)

Selection can be used in combination with: On, Off, Release, Remove, Park, UnPark, Stomp, etc.

SelfFix {Select Select} = Group (Object/Target)

i.e. "SelfFix Executor 8" = "Group Sub 8"

Default {MA + dot} or {MA + .} = Home

**Unfortunately, executing Default as a command by itself also applies to Intensity/Dimmer, which will usually have a value of 0... Whereas Eos Home will only apply to All NIPs by default.

However, "Default (Tap Position Preset Encoder button)" = "Select Last Focus Home"

"Extract" {MA + At + At} is like Make Absolute.

But the command "Extract" is used in place of the command "At" in syntax.

i.e. "Extract Preset 4.2" is essentially like "... At Color Palette 2 Make Absolute"

Meaning that gMA2 will apply the values of Color Preset 2 to the currently selected fixtures as hard, unreferenced data.

Path is Curve

Remove {Delete Delete} is like "@ Enter"

i.e. "Remove Fixture 204" approximately = "Chan 204 At Enter"

HOWEVER once you give an attribute a "Remove" value, in MA2 you must Store Merge or Update to the record target in order to actually remove the move instruction, letting tracking values through. Or in the case of Presets, removing the data altogether.

Similar to **Eos** **except** if in **Eos Blind**, "**@ Enter**" takes effect immediately. Whereas in **MA2 Blind**, a **Store Merge** or **Update** is still required.

Delete can also be used to remove values, which is **NOT** possible in Eos
i.e. "**Delete Cue 4 If Fixture 201**" will remove any values for Fixture 201 stored in Cue 4

Release {**Delete Delete Delete**} = "**Release**" found under "**Make Null**" in **Blind** softkeys
i.e. "**Release Fixture 204**" = "**Chan 204** {**Make Null, Release**}"

The same **Store Merge** and **Update** rules apply as mentioned above with **Remove**

Highlight in MA works pretty much the same as **Highlight** in Eos. Though there is **no** such thing as a **Lowlight** level in **MA**. MA only uses the current background value for the non-highlighted fixtures, it does not force a new value on the non-highlighted fixtures.

Solo is like **Remdim**

Highlight + Solo = **Highlight Remdim**

Time is similar to **Time**, but with different syntax.

Time in MA2 by default will alter the individual discrete timing of whatever fixtures are selected.

Time in Eos by default will alter the cue time of whatever Cue is currently selected.

"**Fixture 201 || Time 3**" or "**Fixture 201 Time 3**" = "**Channel 201 Time 3**"

(A **Store** or **Record** command is then required)

To change time of Cue:

"**Assign Cue 2 Time 3**" = "**Cue 2 || Time 3**" or "**Cue 2 Time 3**"

PARK

Park {Pause Pause} is mostly similar to Eos Park...

"Park Fixture 34" = "Channel 34 Park"

"Park Fixture 34 If PresetType "Color"" = "Channel 34 Color Park"

However, there are a few differences/limitations as well...

On MA2, including a specific value within the Park command does **NOT** behave the same way as Eos.

MA2 Park will always assume the entire fixture unless you specify a specific Attribute, Feature, or PresetType.

Whereas Eos will assume Intensity only, unless otherwise specified.

"Park Fixture 34 Attribute "Dim" At 75" = "Channel 34 At 75 Park"

"Park Fixture 34 At 75" actually parks ALL attributes at a value of 75. Not that useful...

In addition, you can only specify hard numeric values within a Park command. Specifying Presets within a Park command does **NOT** work.

"Park Fixture 34 At Preset 4.17" will **NOT** work as expected. Workaround:

"Fixture 34 At Preset 4.17 || Park Fixture 34 If PresetType "Color"" = "Chan 34 Color Palette 17 Park"

UnPark {Go+ Go+} takes a fixture out of Park.

"UnPark Fixture 34" = "Chan 34 Park" (If already Parked)

You can also Park DMX/Addresses. This works the same as Eos.

"Park DMX 3.50 At 30" = "Address 3/50 At 30 Park"

******Some MA2 programmers actually prefer to create a special Super Priority "Park" Executor/Sequence that they can store and update values into, either manually or using macros. Some find this easier to quickly change, turn on/off, rather than working with the sometimes inconvenient syntax of Park.

CLONE

Clone {**Copy Copy**} is an extremely powerful command in MA2, and it is relatable to the **Eos**, though it may seem somewhat backwards from how one performs the same action on Eos.

Clone is essentially the same as going into **Eos Blind**, and using **Recall From** or **Copy To** to copy values across multiple fixtures and objects/targets. So for example:

"Clone Fixture 31 At Fixture 44 If Preset 4.1 thru 4.30" = "Blind || Color Palette 1 Thru 30 || Channel 44 Recall From Chan 31"

"Clone Fixture 4 At Group 12 If Cue 1 Thru 12" = "Blind || Cue 1 Thru 12 || Group 12 Recall From Chan 4"

You can also use a graphical interface for cloning multiple targets/objects at once using the **Clone Interface** under **Setup**.

You can also Clone/Copy all show data at once:

"Clone Fixture 31 At Fixture 44" = "Patch || Chan 31 Copy To Chan 44 {Show Only}"

****A note on Cloning Presets:** By default, Cloning one Fixture to another Fixture will always create **Selective Preset** data, regardless of whether the source data was Universal, Global, or Selective. To **Clone Global Preset** data, you must clone to a **FixtureType {MA + Fixture}**.

i.e. "Clone Fixture 34 At FixtureType 4.1.1 If Preset 4.1 Thru" creates Global data.

i.e. "Clone FixtureType 1.1.1 At FixtureType 4.1.1 If Preset 4.1 Thru" creates Global data, as long as the source data is also already Global.

http://help2.malighting.com/Page/grandMA2/clone_clone_presets/en/3.3/clone

EFFECTS

Effects are probably THE biggest difference between MA and Eos, so it can be hard to compare.

All MA Effects are most essentially equivalent to Eos Linear Effects.

MA does not contain any sort of step-based effects engine (meaning there is no direct equivalent to an Eos step-based effect or an absolute effect)

All MA Effects, like Eos Linear Effects, use a graph to determine the movement of a value. In this regard, an MA effect can **only** contain a High and Low value.

This is roughly the equivalent of saying that all MA effects are limited to only two “steps”.

These High and Low values (two steps) can either be relative or absolute. Meaning they can be plus or minus the background value, or they can be hard absolute values – either a numeric value, or referencing a Preset. (It is NOT possible to set one of the steps to a “Background” value, as you can on the Eos.)

The only way to create the equivalent of a multi-step (more than 2 step) Eos Absolute effect is to create a Chaser (see Chaser in Executor options section).

As far as **channel selection** for effects, an effect in MA can either be a Template effect or a Selective effect.

So roughly:

Template Effect $\cong$ sort of Absolute Effect - only in that it **does not** contain a channel selection and is instead applied to a selection in the programmer/manually.

Selective Effect $\cong$ sort of Step Based Effect - only in that it **does** contain a selection and will only run on that specific channel selection.

In MA, you can create an effect either by directly modifying values in the effect layer of the programmer/fixture sheet/encoders, and then storing to the effect pool,

OR by saying “Edit Effect ##” which gives you the effects editor screen (I find this to be much easier than directly modifying effect values in the programmer)

Additionally, you’ll find that many of the predefined effects will accomplish a lot of what you need, so it’s usually easier to start with those.

Phase is similar to Grouping

Width is similar to Trail

Groups, Blocks, and Wings are essentially shortcuts to changing the Phase values across fixtures

High and Low values are roughly equivalent of setting values in an Absolute Effect. Can also change the setting to be Center and Size mode, which is much more similar to Scale in a Linear effect.

MORE ON UNDERSTANDING EFFECTS

Like I said, Effects are probably one of the more difficult things to equate between MA2 and Eos.

The most important thing to try and understand is that MA's effects engine is an attribute-based effects engine, whereas Eos is a step-based effects engine (yes, even absolute effects have "steps")

What this means is that on MA, effects (and all of the settings that change an effects look) are additional values that belong to each attribute within a fixture or channel, just like the values for RGB, Pan + Tilt, Dimmer, etc.

In the Fixture Sheet, there is a layer view that you can select for "Effects" (Purple colored tabs on the bottom). This changes the view to show all of the values for effects for those fixture attributes.

This is ultimately where all effect information is applied and stored, **just like any other attribute value such as intensity or RGB values.**

On Eos, effects are separate "programs" that contain all of the settings to make the effect. The effect is then applied **to** channels, and from that point on that effect program is controlling that channel's parameters as an external entity.

The way that I like to think about it is MA effects run from within the channels themselves - you can make a single channel flash just simply by manipulating the various effect values on the encoders, without ever calling up an effect #. All the effect values are part of the channels attributes.

Whereas in Eos, an effect is external program that is then applied TO a channel's attributes.

Non-Similar Commands & Syntax - Important commands on MA2:

BLIND & PREVIEW

MA2 has several "Blind" modes called **Blind**, **BlindEdit**, and **Preview**.

Blind is not exactly the same as in **Eos**. The equivalent of **Eos Blind** is a combination of **Preview** and **BlindEdit**, depending on what type of target you want to view.

Preview = **Blind** for **Cues**

BlindEdit = **Blind** for **Presets** and other record targets

Regular **Blind** (Press & release Blind key) utilizes the same programmer as in live, **which means** that anything you may have in your programmer in live will get knocked out when you enter standard **Blind** mode.

BlindEdit (Press & hold Blind key) is a separate Blind programmer, meaning live programmer values will remain. In general, I always prefer using **BlindEdit**, as it is much more similar to Eos blind.

By default on **MA2**, any editing of values in **Blind** or **Preview** requires an **Update** or **Store** command to complete the process. (**Preview** has an option for Auto-Update)
Whereas any changes or edits made in **Eos Blind** are automatic and immediate.

EDIT

Edit is a very common action command for editing and updating contents of almost any kind of target or object, such as **cues**, **presets**, **macros**, **etc.**

i.e. "Edit Macro 23" , "Edit Cue 4" , "Edit Preset 2.9"

"Edit [Record Target] " is partially similar to going into Eos **Blind** and viewing the contents of objects (**cues**, **palettes**, **presets**).

However, while **Edit** can be used in MA2 **Blind**, it also works in the **Live Programmer**.

Meaning that "Edit Preset 2.9" not in Blind will call all the contents of Positions Preset 9 into the Live Programmer, sending live data to all DMX outputs. From here you can easily make changes, then press the flashing **Update** button to update all changed data to that Preset.

Edit can make quick work of updating presets.

Edit is also a common command for many other objects that don't involve attribute data, such as Macros, or even the Patch of a fixture i.e. "**Edit Fixture 201**" opens window popup showing patch info for Fixture 201

Edit can also be used the same way in MA2 **Blind** and **BlindEdit**, which will NOT send to the Live DMX output. See the section on "**Blind**"

ASSIGN

Assign - no simple Eos equivalent. But is a common action-command to change parameters of objects, as well as Map objects to layouts or physical buttons. Can:

- Assign Cues/Sequences to Executors
- Change Timing of Cues, as well as other Cue options (MIB, etc.)
- Assign Macros or Views to specific buttons or soft keys
- Access or modify many options for most object and targets

"**Assign executor 1.14**" opens Executor options.

"**Assign Macro 1 At Viewbutton 1.1**".

"**Assign Sequence 12 At Executor 1.1**" = "**Fader 1/1 {Mapped To} Cue 12/**"

"**Assign Cue 12 {Time} 2.5**" = "**Cue 12 Time 2.5**"

List of many possible attribute options using Assign:

<https://support.actlighting.com/knowledgeBase/11662799>

STATUS

Status is a term/command used in various instances to include tracked values.

MA2 by default, when pulling data or copying data from other **Cues**, will only look at hard stored values. It will **NOT** include any tracked values unless the **/status** keyword is included in the command.

Eos by default will always include both hard values and tracked values in any **Recall From** or **Copy To** command.

"**Fixture 203 At /status Cue 34**" **truly** = "**Chan 203 Recall From Cue 34**"

"**Copy Cue 4 At Cue 10 /status**" **truly** = "**Cue 4 Copy To Cue 10**"

There is no such thing as a "**Submaster**" in MA2. Only **Sequences** (**cue lists** in Eos terms). To create the equivalent of a "**Submaster**" you simply Store a **Sequence** with one **Cue**.

There isn't really a way to "Select Last" in MA2. But you're less likely to need such a command because MA2 keeps your channel selection until you tell it not to. So you're less likely to lose your channel selection accidentally.

EXECUTOR/FADER OPTIONS

Accessed by Command "Assign Executor #.#" or tapping on top title bar of executor label i.e. "Assign Executor #.#" = "Sub # {Properties}" or "Cue #/ {Properties}"

Here you can find a lot of options for things like:

- Tracking vs cue Only
- Priorities
- MIB settings (Marking)

Flash button is like your standard bump button.

Swop is like "Solo" (as of Eos v2.5)

Black is like a Blackout button for that specific executor

Off on Overwritten is essentially the same as the Off on Stomp

Temp Fader will make an Executor's fader behave much like a default Submaster In that all parameters will fade with the fader, as opposed to just intensity parameters.

Group master is like an Inhibitive Submaster

Chaser turns the Sequence/Executor into something sort of resembling an Absolute or Step-based effect. When Chaser is enabled, all of the cues in the sequence automatically loop based on the cue's time. The Cues essentially become the steps of the effect.

MA2 Priority is similar to Eos Priority, but MA2 Priority also includes LTP/HTP behavior as part of different priorities, whereas Eos has a separate option for LTP vs HTP.

*** It is important to note that all MA2 executors default to LTP, whereas all Eos Faders default to HTP. In MA2, the LTP/HTP option is within the Priority selection (there is no specific separate LTP/HTP option, as there is in Eos).

External Resources

<https://support.actlighting.com/knowledgeBase>